

1. Artifact Evaluation

This section outlines the steps required to recreate the plots presented in the paper. It is worth noting that the safety verification method proposed in this work is inherently *statistical*. Consequently, the reproduction of the plots does not yield an exact match but rather a *stochastic recreation*. In other words, the plots are recreated using the same set of parameters as those reported in the paper, which define the underlying distributions. Each recreated plot therefore represents one possible outcome sampled from the same distribution, hence the term “stochastic recreation”. As a result, some figures that were initially inferred to be safe in the paper may yield unsafe results (or vice versa) during recreation.

This behavior does not affect the soundness of the proposed method. Instead, it reaffirms the stochastic nature of the approach and highlights the influence of various parameters on the safety verification process, as explored in the research questions of the paper. While the recreated plots may not be identical in a deterministic sense, they are stochastically equivalent.

The primary sources of stochasticity in the recreated plots are as follows:

- Although identical parameters (logging probability and log noise) are used, the generated logs themselves are random but stochastically equivalent, as they are sampled from the same distribution.
- The safety verification procedure is inherently stochastic, and therefore safety inferences may vary across different executions.

Recreating Figures

Figures A.2(a)–(c), A.3(a)–(d), A.4(a)–(d), and B.5.. To recreate Figs. A.2(a)–(c), A.3(a)–(d), A.4(a)–(d), and B.5, perform the following steps:

1. Navigate to the Posto repository:

```
cd /path/to/Posto
```

2. If the repository location is not already exported, set the environment variable:

```
export POSTO_ROOT_DIR=/path/to/Posto
```

3. Run the evaluation script with the desired figure identifier:

```
python artEval.py --fig=A2a
python artEval.py --fig=A2b
python artEval.py --fig=A2c

# Choose from A3a, A3b, A3c, A3d
python artEval.py --fig=A3a

# Choose from A4a, A4b, A4c, A4d
python artEval.py --fig=A4a
```

Note: Executing the command for Fig. A.4 first generates a plot for the auxiliary state variable, followed by the target figure. Closing the first figure window displays the next plot. This behavior is consistent across Figs. A.4(a)–(d).

```
python artEval.py --fig=B5
```

Figures B.7(a)–(d)... To recreate Figs. B.7(a)–(d), perform the following steps:

1. Navigate to the Posto repository:

```
cd /path/to/Posto
```

2. If required, export the repository path:

```
export POSTO_ROOT_DIR=/path/to/Posto
```

3. Execute the neural-network-based evaluation script with the desired figure identifier:

```
# Choose from B6a, B6b, B6c, B6d
python artEvalNN.py --fig=B6a
```